Statics $\sum F_x = \sum F_y = 0$, $\sum \tau = 0$. $\tau = rF_\perp = r_\perp F$. Pivot at the force you DON'T want ($r=0$). Uniform beam: weight at $L/2$. CCW +.

Sofa / beam (A at 0, B at x_B) $x_B F_B = \frac{L}{2} Mg + \sum x_i m_i g$; $F_A = W_{tot} - F_B$. Ends, no load: $\frac{1}{2} Mg$ each.

Diving board A at end, B at a: $B = \frac{L/2}{a} Mg$ (up), $A = B - Mg$ (DOWN). (L=10, a=1: $5Mg$, $4Mg$.)

Seesaw $x = \frac{L m_k}{2 M_T}$ from center. Seesaw mass: don't care.

Wall beam + cable (θ cable–beam) $x_{max} = \frac{L}{mg}\left(T\sin\theta - \frac{Mg}{2}\right)$.

Ladder (θ from floor, wall frictionless; M man, m ladder) $x_{max} = \frac{L}{M}\left(\mu_s (M+m)\tan\theta - \frac{m}{2}\right)$ **MAX: round DOWN.**

Empty: $\tan\theta_{min} = \frac{1}{2\mu_s}$. To top: $\tan\theta_{min} = \frac{M + m/2}{(M+m)\mu_s}$.

Climb higher: increase θ or μ_s.

Two cords (θ from CEILING) $T = \frac{mg}{\sin\theta}$.

Young's modulus $Y = \frac{F/A}{\Delta\ell/\ell}$: $\Delta\ell = \frac{F\ell}{AY}$, $F = mg$.

Wire: $A = \pi(\frac{d}{2})^2$ (radius = HALF the diameter).

Block: A = top face (width × length), ℓ = height.

Constants $G=6.67\times10^{-11}$, $M_E=5.98\times10^{24}$, $R_E=6.37\times10^{6}$, $M_{sun}=2\times10^{30}$, 1 AU $=1.5\times10^{11}$ m, $c=3\times10^{8}$, 1 day $=86{,}400$ s.

g at height $g=\frac{GM}{r^2}$, $r=R+h$ (from CENTER, not h). Person's mass cancels.

Orbit period $T=2\pi\sqrt{\frac{r^3}{GM}}$, $r=R+h$. Satellite mass cancels. LEO 400 km: 92.4 min.

Altitude from T $r=\left(\frac{T^2GM}{4\pi^2}\right)^{1/3}$, $h=r-R$ (T in s). Geo: 24 h, $h=35{,}900$ km.

Binary stars $T^2=\frac{16\pi^2r^3}{GM}$, $r=\frac{d}{2}$ (separation $=2r$), $M=$ one star.

Rocket $U=-\frac{GMm}{r}$ (NOT mgh).

Speed for height: $v=\sqrt{2GM\left(\frac{1}{R}-\frac{1}{r}\right)}$, $r=R+h$.

Height from speed: $h=\frac{2GMR}{2GM-v^2R}-R$ (keep 4+ sig figs).

Escape $v_{esc}=\sqrt{\frac{2GM}{R}}$ (Earth 11.2 km/s).

Event horizon (v=c): $r=\frac{2GM}{c^2}$.

Planet speed $v=\sqrt{\frac{GM_{sun}}{r}}$, $GM_{sun}=1.33\times10^{20}$; r in m (AU$\times1.5\times10^{11}$).

Equal pull $\frac{x}{d}=\frac{k}{1+k}$, $k=\sqrt{\frac{M_E}{M_b}}$, x from Earth.

Choices Nearest choice; MAX ("safely") $\rightarrow$ largest that works; MIN $\rightarrow$ smallest that works.